

Sushil Dalavi

sushildalavi.dev — github.com/sushildalavi — sdalavi@usc.edu — +1 (213) 691-3794 — linkedin.com/in/sushildalavi

EDUCATION

University of Southern California

Master of Science in Computer Science | GPA: 3.5/4.0

Coursework: Analysis of Algorithms, Database Systems, Information Retrieval, Applied Cryptography

Aug 2024 – May 2026

Los Angeles, CA

University of Mumbai

Bachelor of Engineering in Computer Engineering | CGPA: 9.71/10.0

Coursework: Data Structures, Operating Systems, Database Management Systems, Distributed Computing

Aug 2019 – May 2023

Mumbai, India

EXPERIENCE

Software Engineer - AI Infrastructure, USC Annenberg Norman Lear Center — Los Angeles, CA Jun 2025 – May 2026

- Delivered **1M+** validated multilingual records across **3** country datasets through a multi-stage **AWS** pipeline (**S3**, **Glue**, **Athena**, **Lambda**), enforcing schema-versioned outputs for UPenn and World Bank collaborators.
- Reduced manual validation workload by **85%** by automating LLM-output validation with a **FastAPI** orchestration layer, OpenAI/Gemini fallback routing, **Pydantic** schema checks, JSON repair, and retry handling.
- Engineered a resilient authenticated **GraphQL** ingestion workflow with **Playwright**, session refresh, structured retries, and parse validation to collect post-level engagement fields across long-running social data pulls.
- Shortened media-to-transcript turnaround by **40%** by splitting sequential audio processing into parallel async stages for download, speech-to-text, speaker diarization, and structured transcript export.

Software Engineer, Reliance Jio Platforms Limited — Mumbai, India

Dec 2023 – Jul 2024

- Cut p99 latency from **2.8s** to **480ms** at **1,500+ RPS** by profiling hot-path queries, adding targeted **MySQL** composite indexes, and isolating high-frequency reads behind a **Redis** look-aside cache.
- Eliminated recurring OOM failures in a **25K+**-file document pipeline by redesigning upload handling into a **Kafka**-orchestrated chunked streaming flow to **AWS S3**, bypassing a hard **10MB** API Gateway payload limit.
- Standardized **Java/Kotlin** service-layer contracts and shared domain models across **6 Spring Boot** microservices, improving consistency across inventory, operations, and service-management workflows.
- De-risked **3** major service upgrades by configuring **NGINX** traffic shadowing and canary rollouts, catching **2** high-severity regressions before customer impact.

TECHNICAL PROJECTS

ReplayForge (GitHub) — Async Workflow Replay Platform

Go, Redis Streams, PostgreSQL, FastAPI, Docker

- Verified crash-safe replay behavior with a **100K**-event chaos run by killing an active Go worker mid-processing and confirming full event convergence with no stream backlog or orphaned records.
- Sustained **3,000+ messages/sec** on an async workflow replay platform using **Redis Streams**, **PostgreSQL**, dead-letter queue, and deterministic replay for backend failure debugging.
- Prevented message loss during worker failures by enforcing **PostgreSQL** claim locking, database-before-ack write ordering, and **Redis Streams** pending-entry recovery.

QueryLens (GitHub) — PostgreSQL Query Observability Platform

C++, Kafka, PostgreSQL/pgvector, Prometheus/Grafana

- Streamed **100K+** PostgreSQL query events from a **C++** collector using **libpqxx** and **pg_stat_statements**, publishing telemetry through **Kafka** into a **FastAPI** control plane and React dashboard.
- Detected **8** deterministic regression classes across relational and **pgvector** workloads, including latency spikes, sequential-scan fallbacks, HNSW bypass, row-estimate mismatch, and temp spills.
- Exercised regression detection across **9** seeded failure scenarios, covering all **8** regression classes with replay-safe event IDs, DLQ routing, and **Prometheus** alerts.

SchemaPilot (GitHub) — Event-Driven API Contract Registry

FastAPI, PostgreSQL, Kafka, Prometheus/Grafana, Docker

- Raised request-path throughput by **8x** and reduced p95 latency by **90%** by replacing synchronous webhook delivery with a **Kafka**-backed transactional outbox and async workers.
- Implemented a runtime API contract monitor that fingerprints live JSON payloads, compares endpoint baselines, and routes schema-change alerts through retry/backoff and DLQ handling.
- Stress-tested contract ingestion across **32K+** requests at **200** VUs, instrumenting **Prometheus** metrics for advisory-lock contention, webhook latency, drift severity, and retry failures.

TECHNICAL SKILLS

Languages: Go, C++, Java, Kotlin, Python, SQL, Bash, TypeScript, JavaScript

Backend & APIs: Spring Boot, FastAPI, REST, GraphQL, gRPC, SQLAlchemy, Hibernate, Pydantic

Databases & Streaming: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Redis Streams, pgvector

Cloud & DevOps: Apache Spark, AWS, GCP, Docker, Kubernetes, Linux, GitHub Actions, CI/CD

Testing, Observability & Frontend: Pytest, JUnit, Playwright, Prometheus, Grafana, React, Vite, Tailwind CSS